Gemini Recommendations from script. Here is a prioritized list of architectural, performance, and maintainability recommendations to take Master List Framework v1.6.14 from a production-ready script to an enterprise-hardened system.

1. High Priority: Scalability & Concurrency

Implement LockService for Multi-User Concurrency

Current State: Multiple users running automated workflows simultaneously could cause race conditions when reading, modifying, and writing back state (e.g., PropertiesService storage for QA runs, Timing Reports, or index generation).

Recommendation: Wrap major entry-point workflows (processDemoP, createMasterList, runDashboardQualityFull) in a document-level lock using Google Apps Script's LockService.

JavaScript

```
const lock = LockService.getDocumentLock();

if (!lock.tryLock(10000)) {

throw new Error("Another user is currently executing a structural workflow. Please try again in a few moments.");

}

try {

// Core workflow execution

} finally {

lock.releaseLock();

}
```

Optimize Memory Footprint in MD5 Hashing (computeStableHash_)

Current State: buildSourceHashForRows_ converts entire 2D datasets into massive strings using JSON.stringify(normalizedRows) before computing an MD5 hash.

Recommendation: If datasets approach 15,000+ rows across 50 columns, allocating a single multi-megabyte string in memory can spike V8 heap allocation. Instead, compute a rolling hash by chunking the dataset into blocks of 1,000 rows or hashing row-by-row sequentially.

2. Medium Priority: Maintainability & Configuration

Consolidate Hardcoded "Magic Numbers" and Arrays

Current State: Functions like styleDashboard_, writeDashboardDefaultsFast_, and validateFrameworkTest9Subheaders_ contain hardcoded column width arrays ([250, 160, 240, 180...]), font sizes (14, 8), and hex colors inside function bodies.

Recommendation: Extract all inline visual arrays, dimensions, and hardcoded style rules into SECTION 1 (CONFIGURATION) under a frozen layout dictionary (e.g., LAYOUT_SCHEMAS). This centralizes visual updates and ensures validation tests match generation routines by referencing the exact same constants.

Resolve or Document Architectural Stubs

Current State: Several functions currently execute empty returns (return 0; or return;):

```
removeDisallowedRawDataWorkingColumns_
hideMasterListSourceColumns_ / showMasterListSourceColumns_
applyMonthlyChangeSubsectionFormat_ / applyMonthlyChangeStandardTopRows_
enforceMasterListFinalDisplay_ / lockMasterListRowHeightsAndClip_
```

Recommendation: * If these represent deprecated logic, prune them completely to reduce cognitive overhead and file payload size.

If these represent planned roadmap features, standardize them with formal runtime warnings or explicit structured comments:

JavaScript

```
function hideMasterListSourceColumns_(sheet) {

// TODO [v1.7.0]: Implement automated column hiding per schema definitions.

return 0;

}
```

3. Low Priority: Observability & Robustness

Upgrade to Google Cloud Logging (console.log)

Current State: The script makes extensive use of Logger.log() for warnings, errors, and timing steps.

Recommendation: Transition internal diagnostic logging from Logger.log() to structured V8 console.log(), console.warn(), and console.error(). This natively pipes telemetry into Google Cloud Operations Suite (Cloud Logging), allowing you to set up automated email/SMS alerts if production execution runtime errors exceed specified thresholds.

Add Automated Rollback / Safe-Staging for Multi-Sheet Updates
Current State: Workflows like updateMasterListForMonth_ execute sequential mutations across multiple sheets (deleting old PMR blocks, inserting new rows, updating primary rows). If an API exception happens mid-execution (e.g., a temporary Google network glitch), the workbook may be left in a partially updated state.
Recommendation: Where feasible, generate output buffers completely in memory or within a temporary hidden staging sheet (__STAGING_MASTER). Only after the staging sheet finishes 100% successful validation should the script swap/rename it with the live production sheet.